# Python Scripting for QA Engineer

## Why do you need scripting as a QA Engineer

- Automate testing activities
- Create tools that help in the quality assurance process
- Automate boring stuff ☺








# Python scripting for QA Engineer

## Why as a QA Engineer I need scripting – Tooling

As a QA engineer in QuestaSim team, sometimes I need to create helper tools to help with the quality assurance process. The tools can be used to

- Make sure that all QA and RnD follow the quality process
- Track test case status across time
- Collect data across the releases in the form of DBs
- Create tools that utilize the data in the DBs to recognize patterns





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## Python as a scripting language

There are many reasons to choose Python among other scripting languages, These reasons can be summarized into the following

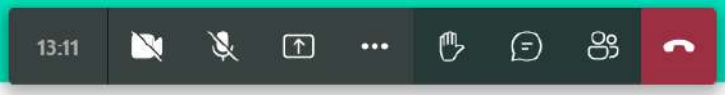
- More powerful
- Easier to learn
- Minimalism
- Community support
- Libraries for almost anything








# Python Basics





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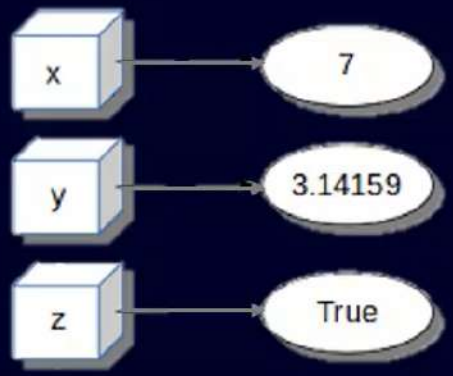
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## Variables

A *variable* is like a box in the computer's memory where you can store a single value. If you want to use the result of an evaluated expression later in your program, you can save it inside a variable.

Python have some naming restrictions. You can name a variable anything if it obeys the following three rules:

- It can be only one word with no spaces.
- It can use only letters, numbers, and the underscore (_) character.
- It can't begin with a number.





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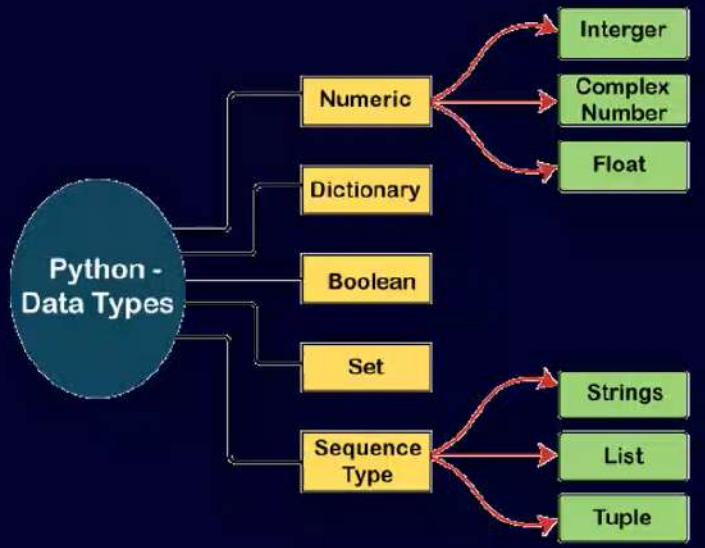
## Primitive Data types

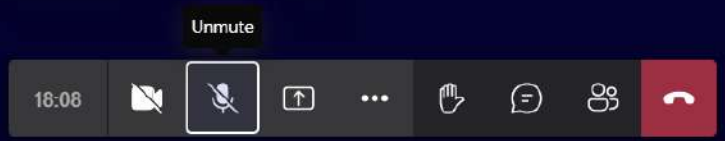
Variables can store data of different types, and different types can do different things.

Python has the following data types built-in by default, in these categories

- Text
- Numeric
- Boolean
- None

To know the data type of a variable type() can be used




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## Arithmetic Operators

| Operator | Operation | Example | Evaluates to |
|---|---|---|---|
| ** | Exponent | 2 ** 3 | 8 |
| % | Remainder | 11 % 3 | 2 |
| / | Division | 10 / 4 | 2.5 |
| * | Multiplication | 2 * 3 | 6 |
| - | Subtraction | 10 - 12 | -2 |
| + | Addition | 1 + 3 | 4 |
| // | Floored division | 10 // 4 | 2 |

**Note: Operator functionality depends on the operands data types**

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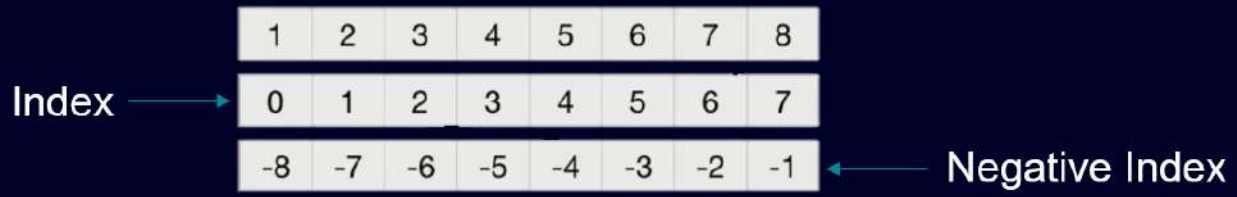
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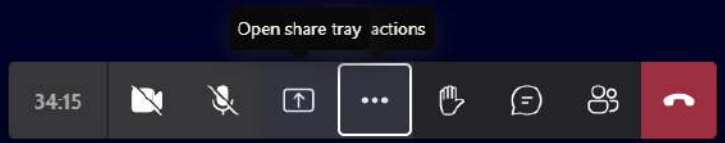
## List

Lists are used to store multiple items in a single variable.

Lists are one of 4 built-in data types in Python used to store collections of data, the other 3 are Tuple, Set, and Dictionary, all with different usage.

- List items are ordered (Retain the same insertion order)
- List items are changeable
- List allow duplicate values
- List items are indexed starting from 0
- List items can be of different types




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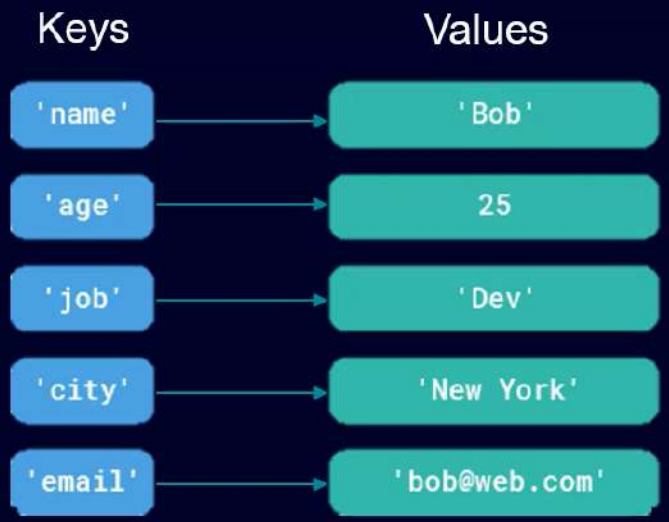

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## Dictionaries

Dictionaries are used to store data values in key: value pairs.

- Dict items are ordered (Retain the same insertion order)
- Dict items are changeable
- Dict allow duplicate values but with different keys
- Dict keys are unique
- Dict keys can be of any (hashable) data type
- Dict values/keys can be of different types







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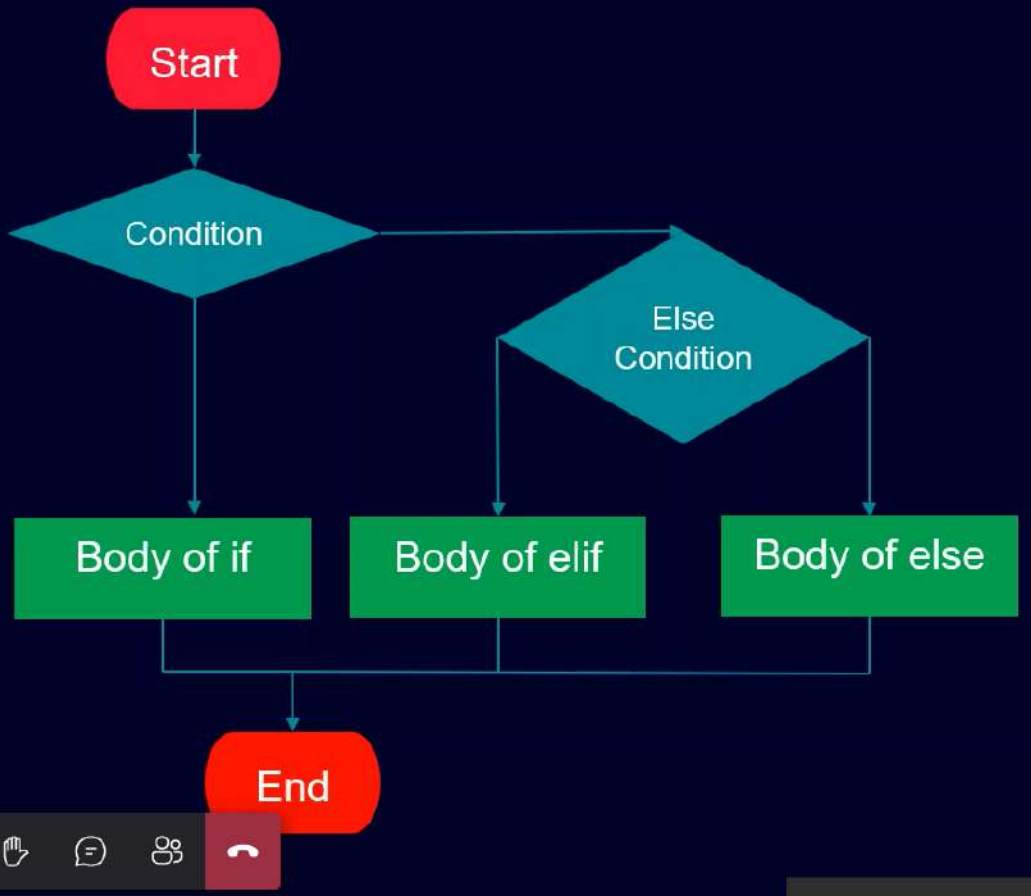

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## Flow Control – If cndition

Flow control statements can decide which Python instructions to execute under which conditions.

- Python supports if conditions for flow control
- If condition can have only one else statement
- If condition can have multiple elif statements
- None type is evaluated to False
- 0 is evaluated to False
- Any number is evaluated to True




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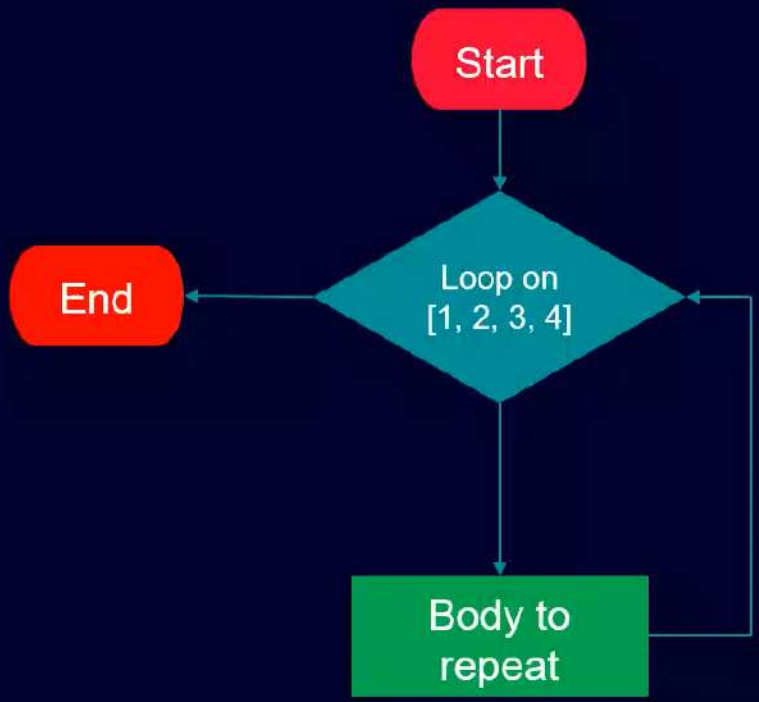
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## Flow Control – For loop

You can make a block of code execute repeatedly using a for loop to iterate over a sequence. Using for loop you can execute a set of statements on each item in the sequence

- break keyword can be used to exit the loop
- continue keyword can be used to skip a loop




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## Functions

A function is a block of code can be used repeatedly in any part of the program

- Functions can have zero or more arguments
- Functions can return zero or more values
- When no return statement is used the function return a None value
- Arguments in functions can have default value
- Functions are used to abstract code (Hide the actual functionality from the function user)
- More complex paradigms can be implemented like functional programming using functions




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## Functions

| Function Name | Related to | Description | Example | Evaluates to |
|---|---|---|---|---|
| len | str, list | Get the size of the string/list | len(“abc”) | 3 |
| replace | str | Replace a string with another string | “abc”.replace(“a”, “g”) | “gbc” |
| strip | str | Remove space characters from the begining/end of the string | “ abc def “.strip() | “abc def” |
| join | str | Join list elements with a separator | “#”.join([1, 2, 3]) | “1#2#3” |
| split | str | Split a string into a list | “12, 34”.split(“,”) | [“12”, “ 34”] |
| append | list | Append an element at the end of the list | [1, 2].append(“4”) | [1, 2, “4”] |





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## Functions

| Function Name | Related to | Description | Example | Evaluates to |
|---|---|---|---|---|
| pop | list | Remove an element from the list | [1, 2, 3, 4].pop() | Output: 4<br>List: [1, 2, 3] |
| sort | list | Sort the list | [1, 4, 2, 7].sort() | Output: None<br>List: [1, 2, 4, 7] |
| reverse | list | Reverse a list | [1, 3, 2, 5].reverse() | Output: None<br>List: [5, 2, 3, 1] |
| keys | dict | Get all the keys in the dict | {“a”: 1, “b”: 2}.keys() | [“a”, “b”] |
| get | dict | Get the value of a key | {“a”: 1, “b”: 2}.get(“a”) | 1 |
| pop | dict | Remove a key from the dict | {“a”: 1, “b”: 2}.pop(“a”) | 1 |

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## File Handles

Python supports creating file handles with three modes

- Read
- Write
- Append

Python supports reading text files through the following functions

- open(FILE_PATH)
- close()
- read()
- readlines()
- write()




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## Modules

A Python module is a file containing Python definitions and statements.
It also makes the code logically organized.

- A module can define functions, classes, and variables.
- A module can also include runnable code.
- Grouping related code into a module makes the code easier to understand and use
- Modules can be imported through the import keyword

An example for Python modules are os module. OS module has functionalities related to the operating system.

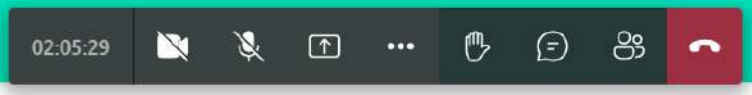



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# Regular Expressions





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## Regular Expressions

If you want to search for a string pattern in a text, you can do that by

- Create a code that iterates on the text chars to match the pattern
- Create a description of the pattern you want to search for and use regex engine to match this pattern




The first option is the naïve one, but it has many drawbacks

- The code is written once for a pattern and can't be reused to match other patterns
- The code structure will be a collection of nested if conditions and loops which is not the best way to do this
- The main advantage of this way is that it doesn't require you to learn new language (regex)

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## Regular Expressions

A regular expression (regex) is a sequence of characters that specifies a search pattern in text. Usually, such patterns are used by string-searching algorithms for "find" or "find and replace" operations on strings, or for input validation.

Regular expressions are used in search engines, in search and replace dialogs of word processors and text editors, in text processing utilities such as sed and AWK, and in lexical analysis. Regular expressions are supported in many programming languages.














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## Regular Expressions - Rules

| Symbol | Description | Symbol | Description |
|---|---|---|---|
| ^ | Matches the beginning of a line | * | Matches zero or more of the preceding element |
| . | Matches any single character except for new lines | ? | Matches zero or one of the preceding elements |
| [ ] | Matches a single char in the group | + | Matches one or more of the preceding elements |
| [^ ] | Matches a single char that is not in the group | \| | Or condition used to match pattern or another pattern |
| $ | Matches the ending of a line | \d | Used to match digits |
| ( ) | Matching a subexpression | \b | Used to match *word boundaries* line, carriage return, spaces) |

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